PROJECT ABSTRACT

On

HOSPITAL MANAGEMENT SYSTEM

A Java-Based Console Prototype

Submitted by

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Abstract

The Hospital Management System is a console-based prototype developed in Java that manages core hospital activities such as patient registration, doctor details, appointment scheduling, prescriptions, and billing in one centralized system. It replaces paper-based record keeping, which is slow and error-prone, with a digital system that gives quick and accurate access to patient information, reducing manual effort and saving time.

The prototype is built with plain Java using object-oriented principles. Separate classes represent the main entities (**Patient, Doctor, Appointment, Prescription, Bill**), and a service class stores them in `ArrayList` collections in memory. A menu-driven interface built with `Scanner` lets the user perform each operation. Validation checks that every appointment and prescription refers to an existing patient and doctor.

The project follows the Waterfall model: requirement analysis, system design, implementation, testing, and maintenance. Each module is tested with sample data to confirm that it works. Future work includes database integration (JDBC with MySQL), a graphical interface, and role-based login.

Technology Stack

Component	Technology
Programming language	Java SE (JDK 8 or above)
Paradigm	Object-oriented programming (classes, encapsulation, constructors, getters)
Data storage	In-memory `ArrayList` collections (java.util)
User interface	Console menu using `Scanner` (java.util)
Development tools	Any Java IDE such as IntelliJ IDEA or Eclipse
Operating system	Platform independent (Windows, Linux, macOS)

Modules

Module	Description
Patient	Register a patient (ID, name, age, gender, phone) and search by ID or name.
Doctor	Add a doctor (ID, name, specialization) and list all doctors.
Appointment	Book an appointment linking a patient and a doctor with a date and time.
Prescription	Record medicine, dosage, and duration for a patient from a doctor. Rejected if the patient or doctor ID does not exist.
Billing	Generate a bill as a fixed consultation fee plus a medicine charge.
Records	View a patient's appointments, prescriptions, and bills together.

Menu Structure

1. Register Patient 2. Add Doctor 3. Book Appointment 4. Add Prescription 5. Generate Bill 6. View Records 7. Exit

Future Scope

- Database integration using JDBC and MySQL for permanent storage
- Graphical user interface using Java Swing or JavaFX
- Role-based login for administrators, doctors, and receptionists
- Report generation and appointment conflict checking